2 Aluminium oxide, Al_2O_3, and phosphorus(V) oxide, P_4O_{10}, are both used as reagents and catalysts.

(a) The melting point of Al_2O_3 is 2072 °C. The melting point of P_4O_{10} is 340 °C.

Explain the difference in the melting points of these two compounds.

.....

.....

.....

..... [3]

(b) A 5.00 dm^3 sealed flask contains 0.400 mol of CO(g) and 0.800 mol of H_2(g) and an Al_2O_3 catalyst. The flask is heated to a temperature of 290 °C and allowed to reach equilibrium. Equation 1 shows the reaction.

equation 1 $\quad CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

The equilibrium constant, K_c, of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

(i) State the units of K_c.

..... [1]

(ii) The equilibrium mixture contains 0.280 mol of CH_3OH(g).

Calculate the value of K_c.

Give your answer to **three** significant figures.

value of K_c = ... [3]

(iii) State and explain the effect, if any, on the value of K_c when the overall pressure in the sealed flask is increased.

.....

..... [1]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(d) In acidic conditions, a dynamic equilibrium is established between $CrO_4^{2-}(aq)$ and $Cr_2O_7^{2-}(aq)$.

$$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l) \qquad \Delta H > 0$$

(i) State what is meant by dynamic equilibrium.

.....

..... [1]

(ii) Identify the condition necessary to establish dynamic equilibrium.

..... [1]

(iii) $CrO_4^{2-}(aq)$ ions are yellow and $Cr_2O_7^{2-}(aq)$ ions are orange.

State what is observed when the following changes are made to an equilibrium mixture of acidified $CrO_4^{2-}(aq)$ and $Cr_2O_7^{2-}(aq)$ ions.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- Dilute HC*l*(aq) is added to the equilibrium mixture.

  observation .....

  explanation .....

  .....

  .....

[4]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) **(i)** Explain why the elements Na to A*l* are good electrical conductors.

.....

..... [1]

(e) $S_2Cl_2(l)$ reacts with $Cl_2(g)$ in a reversible reaction to form $SCl_2(l)$. Under certain conditions, a dynamic equilibrium is established.

$$S_2Cl_2(l) + Cl_2(g) \rightleftharpoons 2SCl_2(l) \qquad \Delta H = -41\,kJ\,mol^{-1}$$

(i) State what is meant by dynamic equilibrium.

.....

..... [1]

(ii) Identify the condition necessary to establish dynamic equilibrium.

..... [1]

(iii) $S_2Cl_2(l)$ is yellow and $SCl_2(l)$ is red.

$$S_2Cl_2(l) + Cl_2(g) \rightleftharpoons 2SCl_2(l) \qquad \Delta H = -41\,kJ\,mol^{-1}$$

State what is observed when the following changes are made to an equilibrium mixture of $S_2Cl_2(l)$ and $SCl_2(l)$.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- The overall pressure of the equilibrium mixture is increased.

  observation .....

  explanation .....

  .....

  .....

[4]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(d) In acidic conditions, a dynamic equilibrium is established between $CrO_4^{2-}(aq)$ and $Cr_2O_7^{2-}(aq)$.

$$2CrO_4^{2-}(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l) \qquad \Delta H > 0$$

(i) State what is meant by dynamic equilibrium.

.....

..... [1]

(ii) Identify the condition necessary to establish dynamic equilibrium.

..... [1]

(iii) $CrO_4^{2-}(aq)$ ions are yellow and $Cr_2O_7^{2-}(aq)$ ions are orange.

State what is observed when the following changes are made to an equilibrium mixture of acidified $CrO_4^{2-}(aq)$ and $Cr_2O_7^{2-}(aq)$ ions.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- Dilute HC*l*(aq) is added to the equilibrium mixture.

  observation .....

  explanation .....

  .....

  .....

[4]

3 $H_2(g)$ and $I_2(g)$ react to form HI(g) in a reversible reaction, as shown in equation 1.

equation 1 $\quad H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

(a) In experiment 1, $H_2(g)$ and $I_2(g)$ are combined.

Water is then added to the equilibrium mixture to produce 1.00 dm^3 of solution **A**.

The amount of $I_2(aq)$ present in 1.00 dm^3 of solution **A** is found by titration with $Na_2S_2O_3(aq)$.

Exactly 32.90 cm^3 of 0.200 mol dm^{-3} $Na_2S_2O_3$ reacts with all of the I_2 in a 25.0 cm^3 sample of solution **A**.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

(i) Calculate the amount, in mol, of $S_2O_3^{2-}$ that reacts in the titration.

amount of $S_2O_3^{2-}$ = ... mol [1]

(b) In experiment 2, 0.100 mol of $H_2(g)$, 0.200 mol of $I_2(g)$ and 0.300 mol of HI(g) are combined.

At equilibrium 0.154 mol of I_2 is present.

Calculate the amounts, in mol, of H_2 and HI in the equilibrium mixture produced in experiment 2.

amount of H_2 in equilibrium mixture = ... mol

amount of HI in equilibrium mixture = ... mol [3]

(c) In experiment 3, 0.0772 mol of H_2 and 0.0986 mol of I_2 are present in an equilibrium mixture at 298 K.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \qquad K_c = 7.94 \times 10^2$$

Use the equilibrium constant, K_c, to calculate the amount, in mol, of HI present in the equilibrium mixture in experiment 3.

amount of HI in equilibrium mixture = ... mol [1]

(d) The value of K_c for the dissociation of HCl at 298 K is 5.50×10^{-34}.

$$2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g) \qquad K_c = 5.50 \times 10^{-34}$$

(i) Define covalent bond.

.....

..... [1]

(ii) Describe and explain the relative thermal stabilities of the hydrogen halides HCl and HI.

.....

.....

..... [2]

(iii) Use the data given in **(c)** and **(d)** to suggest a value for the equilibrium constant for the **dissociation** of HBr(g) at 298 K.

..... [1]

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water. State the formula of the oxide.

..... [1]

(d) State why ZnO is described as a Brønsted–Lowry base when it is added to H_2SO_4(aq).

..... [1]

3 **(a)** Hexene reacts with hydrogen gas to produce hexane.

reaction 1 $C_6H_{12}(l) + H_2(g) \rightarrow C_6H_{14}(l)$

Fig. 3.1 shows the distribution of energies of $H_2(g)$ molecules at temperature *T*.

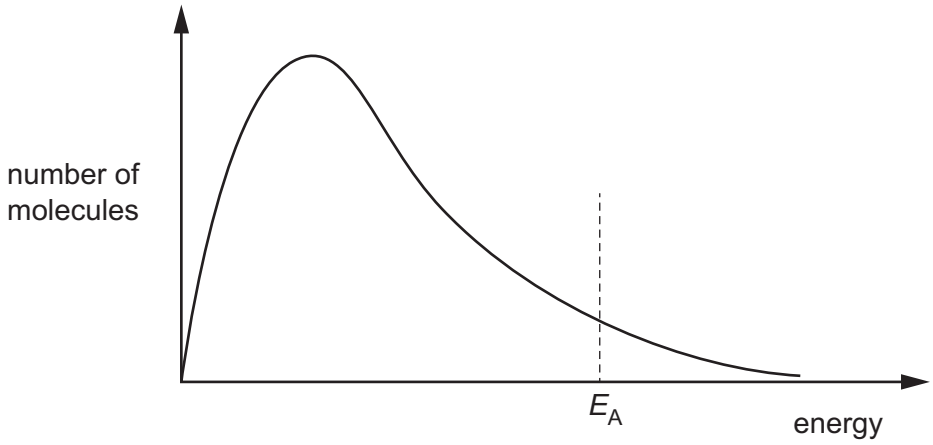


Fig. 3.1

(i) Sketch on Fig. 3.1 the shape of the curve for the same sample of $H_2(g)$ molecules when the temperature is increased. [2]

(b) Define Le Chatelier's principle.

.....

.....

..... [2]

(c) Reaction 2 shows the equilibrium reaction between X(g) and Y(g) to produce Z(g) in a sealed container.

reaction 2 $\quad a\text{X(g)} + b\text{Y(g)} \rightleftharpoons c\text{Z(g)}$

Fig. 3.3 shows the effect of changing pressure on the percentage yield of Z(g) at two different temperatures, 300 K and 350 K.

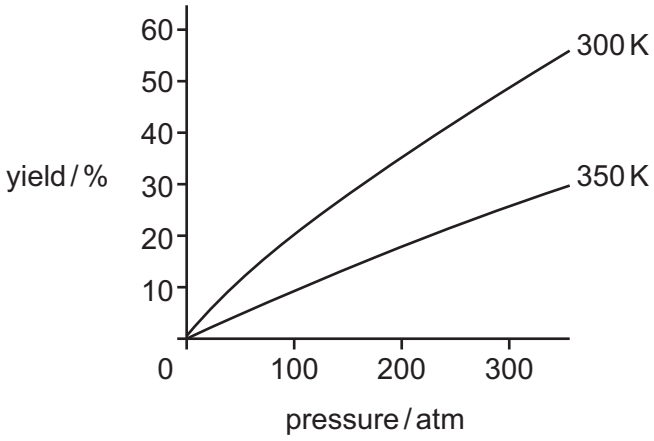


Fig. 3.3

Deduce **two** conclusions about reaction 2 using Fig 3.3.

.....

.....

..... [2]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(ii) In a closed system, PCl_3 and PCl_5 exist in an equilibrium mixture as shown in reaction 1.

reaction 1 $PCl_3(l) + Cl_2(g) \rightleftharpoons PCl_5(s)$ $\Delta H = -124\,kJ\,mol^{-1}$

Deduce **two** conditions that favour the production of PCl_5 in reaction 1.

1 .....

2 .....

[2]

3 The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaC*l*	NaBr	NaI

(a) (i) Complete Table 3.1. [1]

(c) The sodium halides in Table 3.1 also show trends in chemical properties.

(i) Identify the sodium halide that reacts with concentrated H_2SO_4 to form H_2S.

..... [1]

(ii) Identify the sodium halide that **only** undergoes a Brønsted–Lowry acid–base reaction with concentrated H_2SO_4.

..... [1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute HC*l*(aq).

..... [1]

(ii) Describe the thermal stability of the carbonates down Group 2.

..... [1]

(iii) Ammonium carbonate undergoes an acid–base reaction with NaOH(aq).

Explain this statement.

.....

.....

..... [2]

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(i) Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural .....

man-made .....
[2]

(b) A common test for nitrates is the reaction with NaOH and A*l*. Equation 1 shows the reaction.

equation 1 $3NO_3^- + 8Al + 5OH^- + 18H_2O \rightarrow 3NH_3 + 8[Al(OH)_4]^-$

(i) Deduce the oxidation state of nitrogen in NO_3^-.

..... [1]

(iii) NH_3 is a basic gas.

Describe how NH_3 is able to act as a base.

.....

..... [1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute HC*l*(aq).

..... [1]

(ii) Describe the thermal stability of the carbonates down Group 2.

..... [1]

2 $N_2(g)$ reacts with $H_2(g)$ in the Haber process, as shown in reaction 1.

reaction 1 $\quad N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -x\ kJ\,mol^{-1}$

Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	**B**	**C**
initial molar ratio of N_2 : H_2 added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $NH_3(g)$ at equilibrium	58	x	y

(a) Describe and explain the change, if any, to the percentage yield of $NH_3(g)$ produced in **B** compared to **A**.

.....

.....

..... [1]

(b) **(i)** Describe and explain the change, if any, to the percentage yield of $NH_3(g)$ produced in **C** compared to **A**.

.....

.....

..... [1]

(c) **(i)** Write an expression for the equilibrium constant, K_p, for reaction 1. State the units.

K_p =

units .....

[2]

(ii) Equilibrium mixture **D** is made when 1.0 mol of $N_2(g)$ and 3.0 mol of $H_2(g)$ are added to a sealed container at 750 °C and 1000 atm and left to reach equilibrium. This mixture contains 1.16 mol of $NH_3(g)$.

Calculate the mole fraction of $NH_3(g)$ in **D**.

mole fraction of $NH_3(g)$ = ... [2]

(iii) The mole fraction of $N_2(g)$ is 0.625 in a new equilibrium mixture, **E**.

Calculate the partial pressure of $N_2(g)$ in **E** when the total pressure is 1000 atm.

partial pressure of $N_2(g)$ = ... atm [1]

4 **(a)** $NH_3(g)$ reacts with $HCl(g)$ to produce $NH_4Cl(s)$, as shown.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl.

[2]

(b) An exothermic reaction occurs when $NH_4^+(aq)$ is added to $OH^-(aq)$.

(i) Identify the type of reaction.

..... [1]

1 **(a)** Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

(e) Two Period 3 elements react with an excess of oxygen at room pressure.

(i) Complete Table 1.2.

Table 1.2

1	2	3
Period 3 element	state of oxide at room temperature and pressure	approximate pH of solution made when oxide is added to water
Na		
S		

[2]

(ii) The solutions made in column 3 of Table 1.2 are mixed together.
Name the type of reaction that occurs.

..... [1]

3 **(a)** Describe what is meant by dynamic equilibrium.

.....

.....

..... [2]

(b) Reaction 4 describes the reversible reaction between yellow $Fe^{3+}(aq)$ and colourless $SCN^{-}(aq)$ to produce red $FeSCN^{2+}(aq)$.

reaction 4 $Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$

yellow colourless red

An equilibrium mixture contains $Fe^{3+}(aq)$, $SCN^{-}(aq)$ and $FeSCN^{2+}(aq)$. A few colourless crystals of soluble KSCN(s) are added. The mixture is then left until it reaches equilibrium again. The temperature of both equilibrium mixtures is the same.

(i) Deduce the changes that occur, if any, in the equilibrium mixture after KSCN(s) is added compared to the original equilibrium mixture.

- change in appearance

  .....

- change in relative concentration of $Fe^{3+}(aq)$

  .....

- change in value of the equilibrium constant, K_c

  .....

[3]

(ii) The expression for the equilibrium constant, K_c, for reaction 4 is shown.

$$K_c = \frac{[FeSCN^{2+}(aq)]}{[Fe^{3+}(aq)] \times [SCN^{-}(aq)]}$$

5.00×10^{-5} mol of $Fe^{3+}(aq)$ and 5.00×10^{-5} mol of $SCN^{-}(aq)$ are added together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3.

At equilibrium the concentration of $FeSCN^{2+}(aq)$ is 4.23×10^{-4} mol dm^{-3}.

Calculate the equilibrium constant, K_c, for reaction 4.

Include the units in your answer.

K_c = ...

units .....

[4]

2 **(a)** Sulfur chloride, SCl_2, is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

..... [1]

(ii) Magnesium nitride reacts with an excess of water to produce ammonia and magnesium hydroxide only. Construct an equation to describe this reaction.

..... [1]

(iii) Explain why the solution produced in the reaction in **(c)(ii)** has a pH greater than 7. Refer to the products of the reaction in your answer.

.....

.....

..... [2]

3 **(a)** Define Le Chatelier's principle.

.....

.....

..... [2]

(b) Reaction 1 describes the reversible reaction between yellow $Fe^{3+}(aq)$ and colourless $SCN^{-}(aq)$ to produce red $FeSCN^{2+}(aq)$.

reaction 1 $Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$ $\Delta H = -x\ kJ\,mol^{-1}$

yellow colourless red

A mixture of $Fe^{3+}(aq)$, $SCN^{-}(aq)$ and $FeSCN^{2+}(aq)$ is at equilibrium at 20 °C.

The temperature of this mixture is then increased to 50 °C and allowed to reach equilibrium.

Deduce the changes that occur, if any, in the equilibrium mixture at 50 °C compared to the equilibrium mixture at 20 °C.

- change in appearance

  .....

- change in relative concentration of $FeSCN^{2+}(aq)$

  .....

- change in value of the equilibrium constant, K_c

  .....

[3]

(c) In another experiment, equimolar amounts of $Fe^{3+}(aq)$ and $SCN^{-}(aq)$ are mixed together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3.

reaction 1 $Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$

At equilibrium the mixture contains:

- $[SCN^{-}] = 1.30 \times 10^{-3}\ mol\,dm^{-3}$
- $[FeSCN^{2+}] = 0.300 \times 10^{-3}\ mol\,dm^{-3}$.

(i) Calculate the initial amount, in mol, of $Fe^{3+}(aq)$ added to $SCN^{-}(aq)$ to produce this mixture.

initial amount of $Fe^{3+}(aq)$ = ... mol [2]

(ii) Calculate K_c for reaction 1 and state its units.

Show your working.

K_c = ...

units .....
[2]

2 NO and NO_2 react at 25 °C to give N_2O_3 as shown in the equation.

$$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g) \qquad \Delta H = -7.2\,kJ\,mol^{-1}$$

The reaction is reversible and reaches equilibrium in a closed system.

(a) Fig. 2.1 shows how the rate of the forward reaction changes with time.

Initially, the rate of the reverse reaction is zero.

Complete Fig. 2.1 to sketch how the rate of the **reverse** reaction changes with time.

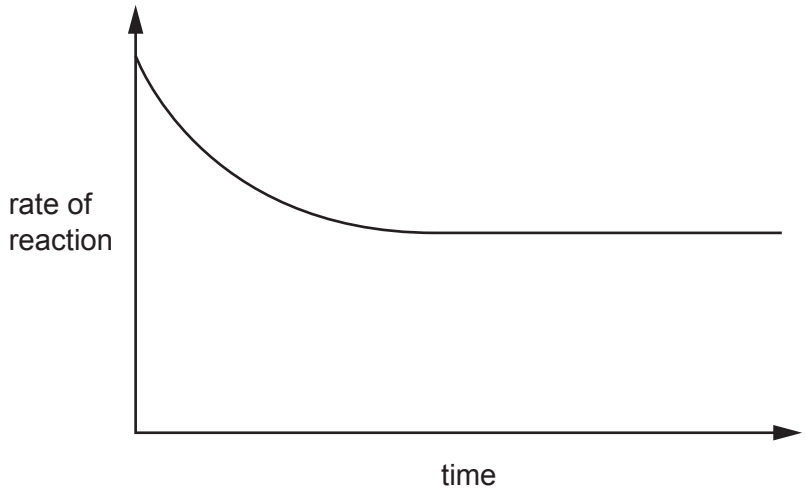


Fig. 2.1

[1]

(b) State how the position of equilibrium changes, if at all, when the reaction takes place at 100 °C.

Explain your answer.

Assume the pressure remains constant.

.....

.....

.....

..... [2]

(c) Table 2.1 shows the composition of an equilibrium mixture of NO(g), NO_2(g) and N_2O_3(g) at 101 kPa.

Table 2.1

gas	number of moles at equilibrium / mol
NO	0.605
NO_2	0.605
N_2O_3	0.390

Calculate K_p, the equilibrium constant with respect to partial pressures.

Deduce the units of K_p.

K_p = ... units .....
[3]

(f) A student titrates nitric acid with a base to form a solution containing aqueous magnesium nitrate.

(i) Identify a base that the student could use.

..... [1]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

[2]

(ii) H_3PO_4 is a weak Brønsted–Lowry acid.

Define weak Brønsted–Lowry acid.

.....

.....

..... [2]

3 A neutralisation reaction occurs when NaOH(aq) is added to H_2SO_4(aq).

equation 1 $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

(a) Define enthalpy change of neutralisation, ΔH_{neut}.

.....

.....

..... [2]

(c) The enthalpy change of neutralisation of CH_3COOH(aq) and NaOH(aq) is $-55.2\,kJ\,mol^{-1}$.

(i) Complete the equation for the reaction.

.....CH_3COOH + ...NaOH $\rightarrow$ ..... [1]

(ii) Values for the enthalpy change of neutralisation, ΔH_{neut}, are shown in Table 3.1.

Table 3.1

reagents	ΔH_{neut} / kJ mol^{-1}
NaOH + HC*l*	–57.1
NaOH + CH_3COOH	–55.2

Suggest why the value for ΔH_{neut} of the weak acid, CH_3COOH, reacting with NaOH is different to the value obtained using the strong acid, HC*l*. Assume that the values are determined under the same conditions.

.....

..... [1]

4 **(a)** Hydrogen chloride gas is made in the laboratory by adding concentrated sulfuric acid to potassium chloride.

(i) Construct an equation for this reaction.

..... [1]

(b) A sample of HI(g) is added to a 2.00 dm^3 sealed vessel at 764 K and allowed to reach equilibrium.

reaction 1 $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ $K_c = 0.0217$ at 764 K

At equilibrium the mixture contains 1.70 mol of HI(g).

(i) State **one** difference in the appearance of the initial reaction mixture compared to the mixture at equilibrium.

.....

..... [1]

(ii) Deduce the expression for equilibrium constant K_c for reaction 1.

K_c =

[1]

(iii) Calculate the concentration of I_2 present in the reaction mixture at equilibrium.
Show your working.

concentration of I_2 = ... mol dm^{-3} [3]

(c) The experiment is repeated at 500 K. The value of K_c under these conditions is 0.00625.

(i) Describe the difference in the composition of the equilibrium mixture at 500 K compared to 764 K.

.....

..... [1]

(ii) Use Le Chatelier's principle to deduce whether the decomposition of HI(g) is endothermic or exothermic. Explain your answer.

.....

.....

..... [1]

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4.

reaction 1 $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

(a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r, for reaction 1.

Table 4.1

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

ΔH_r = ... kJ mol^{-1} [2]

(b) Reaction 1 reaches equilibrium at constant temperature and pressure.

Deduce what effect increasing the pressure will have on the amount of ethanol in the new equilibrium mixture. Use Le Chatelier's principle to explain your answer.

effect of increasing pressure .....

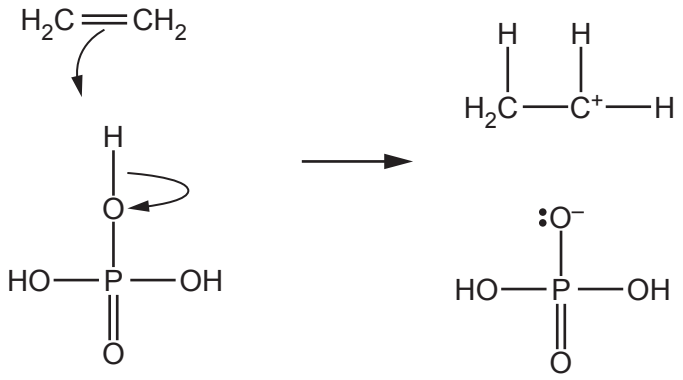
explanation .....

.....

[2]

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

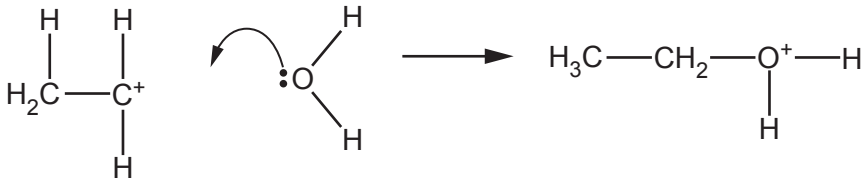


Fig. 4.1

(i) Describe the behaviour of H_3PO_4 in step 1 in Fig. 4.1. Explain your answer.

.....

..... [1]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

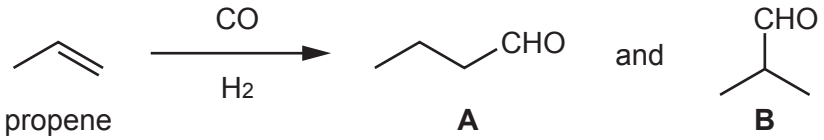


Fig. 3.1

(a) **(i)** Define addition reaction.

.....

..... [1]

(d) The reaction of ethene, C_2H_4, with a 1:1 mixture of CO and H_2 is shown in equation 1.

equation 1 $C_2H_4(g) + CO(g) + H_2(g) \rightleftharpoons CH_3CH_2CHO(g)$

At atmospheric pressure a cobalt-based catalyst is used in this reaction.

(i) State and explain the effect of using a catalyst on this reaction.

.....

.....

.....

..... [2]

(ii) Explain why the yield of $CH_3CH_2CHO(g)$ increases when the overall pressure of the reaction mixture is increased.

.....

..... [1]